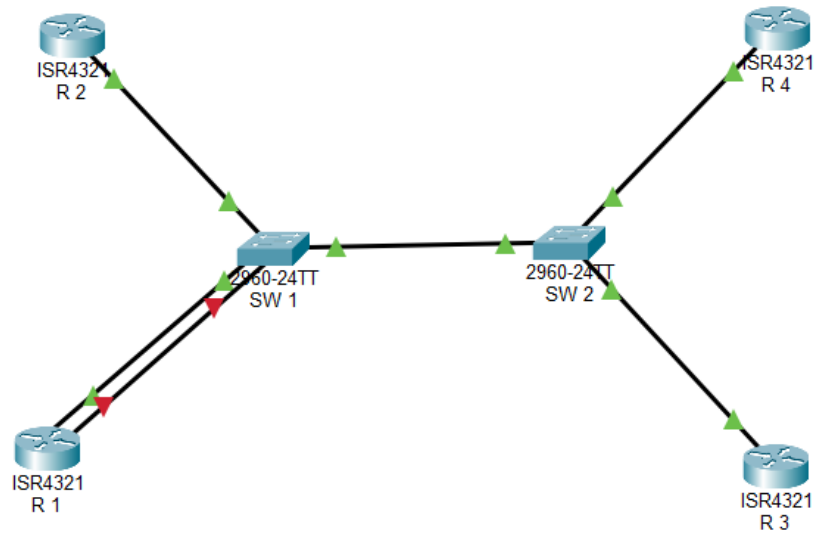


Week 2: Cisco Device Functions (5 Marks)



Task 5:

SW1 :

```
Switch>ena
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.4297.2c18   DYNAMIC   Fa0/24
1       000a.f3a7.d001   DYNAMIC   Fa0/2
1       0060.2f2d.bb01   DYNAMIC   Fa0/1
Switch#
```

SW2:

```
Switch>ena
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000a.41c1.2a02    DYNAMIC   Fa0/3
1       000c.cf82.6201    DYNAMIC   Fa0/4
1       00d0.ba61.8c18    DYNAMIC   Fa0/24
Switch#
```

Copy Paste

Task 7:

R1:

R 1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0/0

GigabitEthernet0/0/1

GigabitEthernet0/0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.2F2D.BB01

IP Configuration

IPv4 Address 192.168.1.21

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#no ip address
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.21 255.255.255.0
Router(config-if)#

R2:

The screenshot shows the configuration window for R2, specifically for the GigabitEthernet0/0/0 interface. The left sidebar has tabs for Physical, Config (selected), CLI, and Attributes. Under the Config tab, there is a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. The INTERFACE category is expanded, showing GigabitEthernet0/0/0 and GigabitEthernet0/0/1. The main area displays the configuration for GigabitEthernet0/0/0. The Port Status is set to On. Link Speed is set to 1000 Mbps, 100 Mbps, 10 Mbps, and Auto (selected). Duplex is set to Half Duplex, Full Duplex (selected), and Auto. The MAC Address is 000A.F3A7.D001. The IP Configuration section shows the IPv4 Address as 192.168.1.21 and the Subnet Mask as 255.255.255.0. The Tx Ring Limit is set to 10.

GigabitEthernet0/0/0	
Port Status	☒ On
Link Speed	☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000A.F3A7.D001
IP Configuration	
IPv4 Address	192.168.1.21
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Equivalent IOS Commands

```
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface GigabitEthernet0/0/0  
Router(config-if)#  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface GigabitEthernet0/0/0  
Router(config-if)#ip address 192.168.1.10 255.255.255.0  
Router(config-if)#  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface GigabitEthernet0/0/0  
Router(config-if)#
```

Top

R3:

The screenshot shows the configuration window for R3, specifically for the GigabitEthernet0/0/1 interface. The left sidebar has tabs for Physical, Config (selected), CLI, and Attributes. Under the Config tab, there is a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. The INTERFACE category is expanded, showing GigabitEthernet0/0/0 and GigabitEthernet0/0/1. The main area displays the configuration for GigabitEthernet0/0/1. The Port Status is set to On. Link Speed is set to 1000 Mbps, 100 Mbps, 10 Mbps, and Auto (selected). Duplex is set to Half Duplex, Full Duplex (selected), and Auto. The MAC Address is 000A.41C1.2A02. The IP Configuration section shows the IPv4 Address as 192.168.1.22 and the Subnet Mask as 255.255.255.0. The Tx Ring Limit is set to 10.

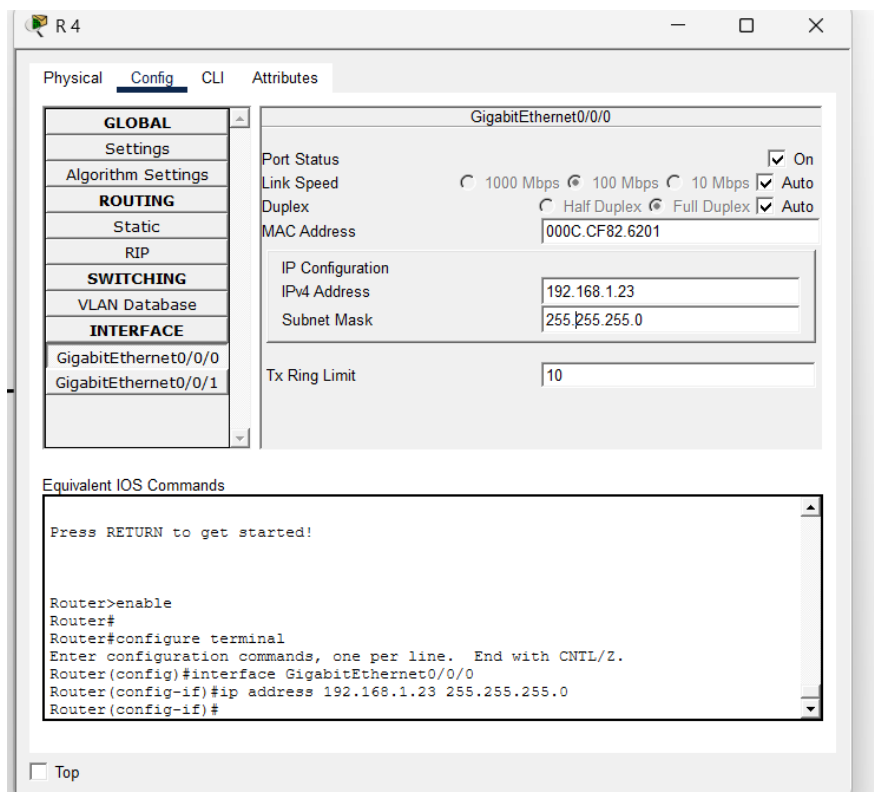
GigabitEthernet0/0/1	
Port Status	☒ On
Link Speed	☐ 1000 Mbps ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	000A.41C1.2A02
IP Configuration	
IPv4 Address	192.168.1.22
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Equivalent IOS Commands

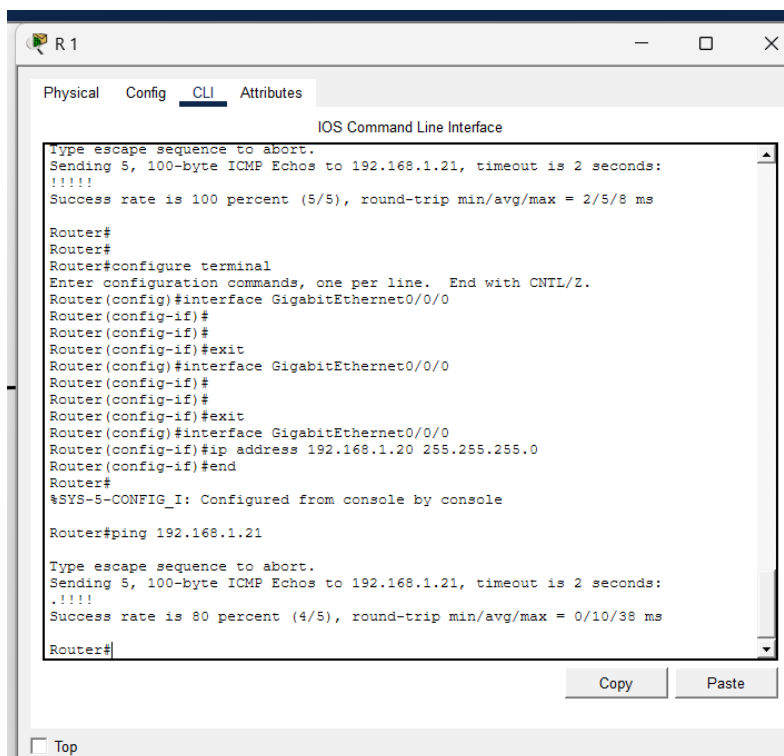
```
Press RETURN to get started!  
  
Router>enable  
Router#  
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#interface GigabitEthernet0/0/1  
Router(config-if)#ip address 192.168.1.22 255.255.255.0  
Router(config-if)#ip address 192.168.1.22 255.255.255.0  
Router(config-if)#
```

Top

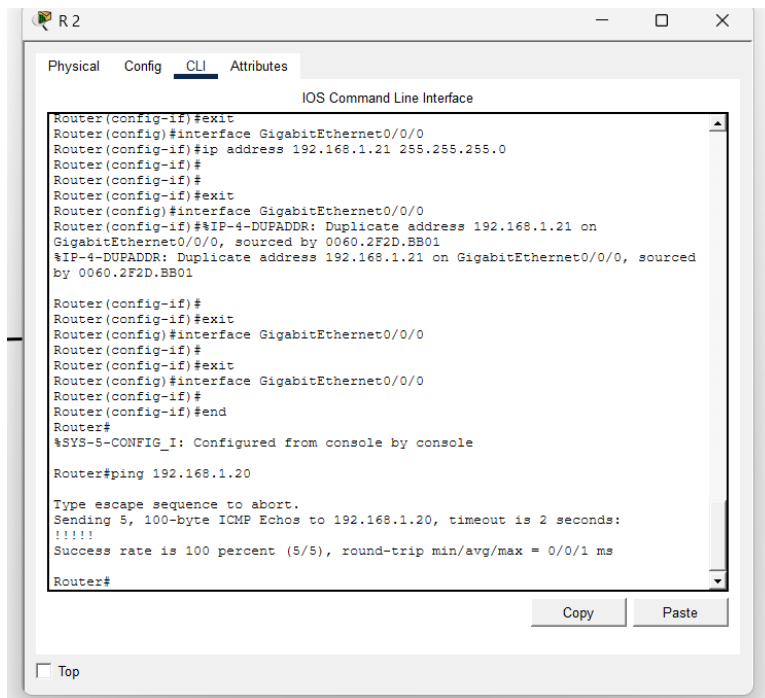
R4:



Ping R1:



Ping R2:



The screenshot shows the CLI window for router R2. The 'CLI' tab is selected. The command history shows the configuration of interface GigabitEthernet0/0/0 with IP address 192.168.1.21. A duplicate address warning is displayed. The configuration is saved, and a ping command is executed to 192.168.1.20. The ping is successful with a 100% success rate.

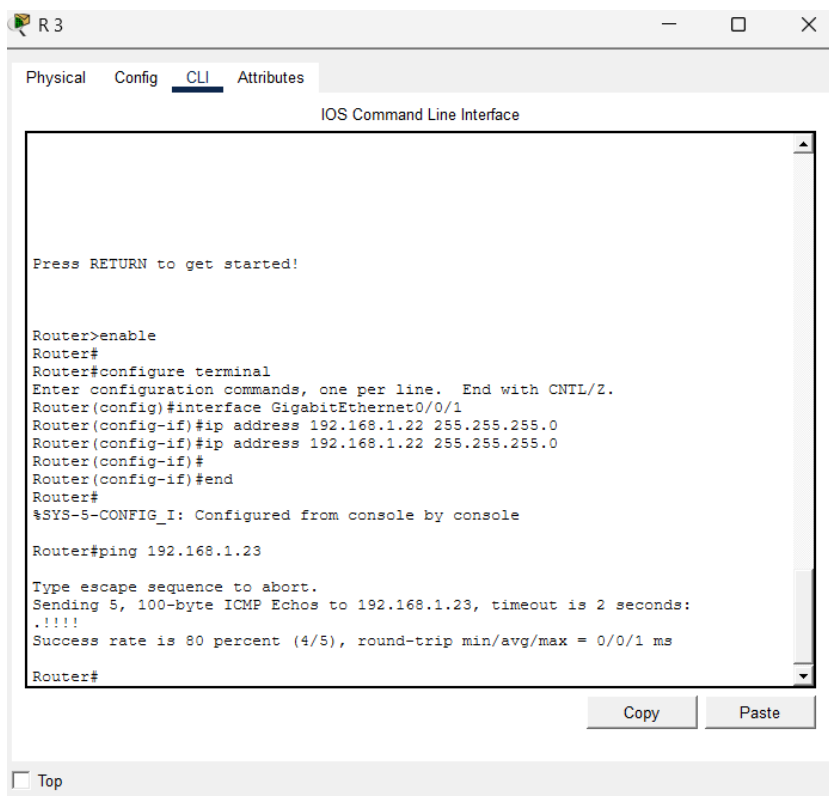
```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.21 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#%IP-4-DUPADDR: Duplicate address 192.168.1.21 on
GigabitEthernet0/0/0, sourced by 0060.2F2D.BB01
%IP-4-DUPADDR: Duplicate address 192.168.1.21 on GigabitEthernet0/0/0, sourced
by 0060.2F2D.BB01
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 192.168.1.20

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.20, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

Ping R3:



The screenshot shows the CLI window for router R3. The 'CLI' tab is selected. The command history shows the configuration of interface GigabitEthernet0/0/1 with IP address 192.168.1.22. A duplicate address warning is displayed. The configuration is saved, and a ping command is executed to 192.168.1.23. The ping is successful with an 80% success rate.

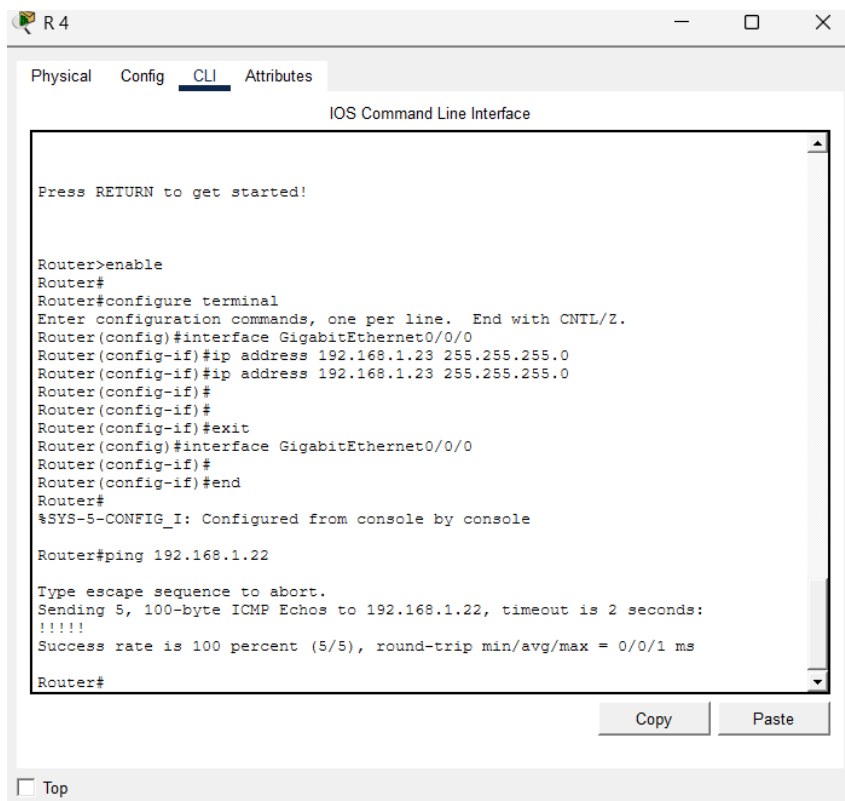
```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 192.168.1.22 255.255.255.0
Router(config-if)#ip address 192.168.1.22 255.255.255.0
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 192.168.1.23

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.23, timeout is 2 seconds:
..!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

Ping R4:



The screenshot shows a terminal window titled 'R4' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The user has entered commands to enable configuration mode, configure the GigabitEthernet0/0/0 interface with IP address 192.168.1.23 and subnet mask 255.255.255.0, and then ping the IP address 192.168.1.22. The ping is successful, showing a 100% success rate.

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.23 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

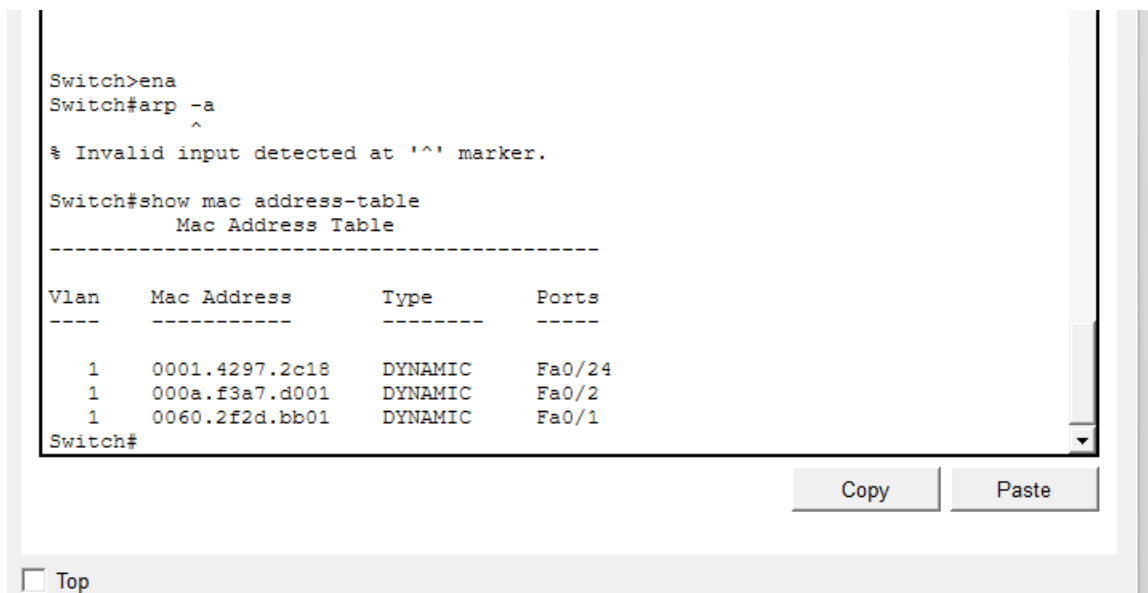
Router#ping 192.168.1.22

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.22, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

Task 8:

SW1



The screenshot shows a terminal window titled 'SW1' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The user has entered commands to enable configuration mode, attempt to configure the ARP table with an invalid input, and then display the MAC address table. The MAC address table shows three entries for VLAN 1, all with dynamic addresses.

```
Switch>ena
Switch#arp -a
^
% Invalid input detected at '^' marker.

Switch#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.4297.2c18    DYNAMIC   Fa0/24
1       000a.f3a7.d001    DYNAMIC   Fa0/2
1       0060.2f2d.bb01    DYNAMIC   Fa0/1

Switch#
```

SW2

```
Switch>ena
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000a.41c1.2a02    DYNAMIC   Fa0/3
1       000c.cf82.6201    DYNAMIC   Fa0/4
1       00d0.ba61.8c18    DYNAMIC   Fa0/24
Switch#
```

Copy

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Task 9:

SW1:

```
Switch>ena
Switch#clear mac address-table dynamic
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
Switch#
```

SW2:

```
Switch>ena
Switch#clear mac address-table dynamic
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       00d0.ba61.8c18    DYNAMIC   Fa0/24
Switch#
```

Task 10:

R1:

```
Router>ena
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.20/32 is directly connected, GigabitEthernet0/0/0

Router#
```

Copy

Paste

R2:

```
Router>ena
Router# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.21/32 is directly connected, GigabitEthernet0/0/0

Router#
```

Copy

Paste

☐ Top

R3:

```
Router>ena
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.22/32 is directly connected, GigabitEthernet0/0/1

Router#
```

Copy

Paste

☐ Top

R4:

```
Router>ena
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.23/32 is directly connected, GigabitEthernet0/0/0

Router#
```

Copy

Paste

☐ Top

Task 11:

R1:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.20 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

Copy

Paste

R2:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 192.168.1.22 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

Copy

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R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.23 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

Copy

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R4:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.21 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

Copy

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Task 12:

R1:

```
Router>ena
Router#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 192.168.1.20    YES manual up
GigabitEthernet0/0/1 unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down
Router#
```

R2:

```
Router>ena
Router#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 192.168.1.21    YES manual up
GigabitEthernet0/0/1 unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down
Router#
```

R3:

```
Router>ena
Router#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0/1 192.168.1.22    YES manual up
Vlan1               unassigned      YES unset  administratively down down
Router#
```

Copy

Paste

R4:

```
Router>ena
Router#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0 192.168.1.23    YES manual up
GigabitEthernet0/0/1 unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down
Router#
```

Copy

Paste

Task 13:

Task 14:

Task 15:

Yes. MAC addresses reappear because the switch dynamically learns them again when it receives traffic from connected devices.